

MUNSTER OSNABRUCK, GermanyWMO: **103150**Lat: **52.1344N**Lon: **7.6969E**Elev: **53**StdP: **100.69**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-5.2	-11.2	1.4	-5.7	-8.9	1.8	-4.5	12.4	8.7	11.1	8.2	2.5	70	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.9	19.5	28.2	18.6	26.1	17.8	20.7	27.2	19.8	26.0	19.0	24.6	3.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	22.6	17.9	12.9	21.8	17.0	12.2	20.9	59.9	27.1	56.6	26.3	53.9	24.7	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.3	-10.3	35.0	3.2	1.8	-12.6	36.3	-14.5	37.3	-16.3	38.4	-18.6	39.7		
(5)			-10.6	22.7	3.2	0.8	-13.0	23.3	-14.8	23.8	-16.7	24.2	-19.0	24.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	2.4	2.7	5.9	10.5	13.6	17.0	19.2	18.1	15.1	11.2	6.7	4.1
	DBStd	6.96	4.34	3.99	4.11	3.98	4.01	3.47	3.50	3.04	3.13	3.55	3.80	4.29
	HDD10.0	952	236	203	134	41	11	1	0	0	1	29	111	186
	HDD18.3	2961	493	437	384	236	154	65	31	42	106	222	349	442
	CDD10.0	1170	1	0	9	56	123	212	286	250	154	65	12	2
	CDD18.3	138	0	0	0	1	9	26	58	33	10	1	0	0
	CDH23.3	1293	0	0	0	23	88	240	567	289	81	6	0	0
	CDH26.7	400	0	0	0	2	17	63	208	92	17	1	0	0
Wind	WSAvg	3.3	3.9	3.6	3.6	3.2	3.2	3.0	3.0	2.8	2.8	2.9	3.3	3.9
Precipitation	PrecAvg	782	71	57	52	45	56	69	79	80	66	63	70	74
	PrecMax	1022	134	122	126	100	130	194	163	257	144	190	134	171
	PrecMin	599	6	15	9	2	23	13	29	23	9	15	0	27
	PrecStd	117	34	30	31	20	26	37	36	50	36	32	33	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.0	19.2	26.1	29.0	31.2	34.9	33.1	29.1	23.8	17.2	14.0
		MCWB	10.6	7.9	11.3	14.9	18.5	19.8	20.4	20.5	18.7	16.9	13.6	11.3
	2%	DB	11.2	11.0	16.2	23.0	25.8	28.2	31.2	29.1	25.8	20.1	15.0	12.2
		MCWB	9.4	8.5	10.4	13.5	16.3	18.9	19.5	19.4	18.0	15.9	12.6	10.1
	5%	DB	9.8	9.8	14.0	20.8	23.2	26.1	29.0	26.1	23.1	18.0	13.2	11.0
		MCWB	8.6	8.0	9.8	13.2	15.0	17.9	18.8	18.2	17.0	14.8	11.3	9.3
	10%	DB	8.1	8.2	12.1	17.9	21.1	23.9	26.2	24.1	20.8	16.2	12.2	9.2
		MCWB	7.1	6.7	8.7	12.0	14.7	17.2	18.1	17.4	16.1	13.7	10.4	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	10.1	12.8	15.8	20.0	21.5	22.0	21.6	20.0	17.4	14.5	11.8
		MCDB	12.5	11.0	16.7	23.1	27.5	28.3	29.7	28.1	26.3	21.6	16.1	13.4
	2%	WB	9.9	8.9	11.4	14.6	17.9	20.2	20.7	20.3	18.7	16.1	13.0	10.7
		MCDB	11.2	10.4	14.9	20.9	23.5	26.2	27.6	26.9	24.3	19.7	14.8	12.2
	5%	WB	8.5	7.8	10.2	13.5	16.4	19.0	19.8	19.2	17.7	15.1	11.9	9.6
		MCDB	9.6	9.5	13.3	19.0	21.4	24.3	26.0	24.9	22.0	17.6	13.6	10.7
	10%	WB	7.1	6.6	9.1	12.5	15.1	17.8	19.0	18.1	16.6	14.1	10.7	8.2
		MCDB	8.0	7.9	11.9	17.3	19.8	22.4	24.9	22.7	20.0	16.2	12.2	9.4
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.5	7.9	10.0	9.8	9.9	10.3	9.7	9.1	7.4	5.4	4.4
		MCDBR	5.1	7.7	11.7	14.1	12.8	13.7	14.6	13.7	12.7	9.9	6.5	5.3
		MCWBR	4.4	5.5	6.8	6.8	6.0	6.2	5.2	5.8	6.2	6.0	4.9	4.4
	5% WB	MCDBR	4.8	6.6	10.3	12.1	11.4	11.3	12.2	11.5	11.2	8.8	6.1	5.1
		MCWBR	4.5	5.3	6.4	6.4	6.1	5.7	5.1	5.5	6.1	5.7	4.9	4.5
Clear-Sky Solar Irradiance	taub	0.324	0.346	0.376	0.410	0.413	0.412	0.424	0.417	0.389	0.360	0.341	0.318	
	taud	2.369	2.344	2.273	2.219	2.245	2.268	2.244	2.284	2.351	2.415	2.382	2.353	
	Ebn at Noon	670	762	808	822	839	843	824	808	787	737	641	606	
	Edh at Noon	61	82	106	127	129	128	129	118	98	76	60	53	
All-Sky Solar Radiation	RadAvg	0.68	1.36	2.47	4.01	4.88	5.21	4.98	4.20	3.05	1.78	0.82	0.53	
	RadStd	0.09	0.24	0.34	0.45	0.45	0.42	0.64	0.36	0.36	0.17	0.10	0.06	